

# CodaDx6 Brief Specification

**Version 1.1.0**

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***General Description***

Chips&Media's CodaDx6 (TristanEx) is an optimally-unified multi-standard video codec IP which performs H.264, MPEG-4, and H.263 decoder, and MPEG-4, H.263, and H.264 encoder up to D1 resolution at 30 frames per second. Under Chips&Media advanced design technologies, CodaDx6 not only requires ultra low-power and ultra low-clock frequency, but also provides decent flexibility in error concealment, error resilient, and multi-resolution/multiplex codec control.

***Standards / Profiles / Levels***

| <i>Supported</i> | <i>Profiles / Levels</i>       |
|------------------|--------------------------------|
| Decoder          | MPEG-4 Simple Profile          |
|                  | H.263 Profile 0 Level 10-50    |
|                  | H.263 Profile 3 Level 10-50    |
|                  | H.264 Baseline Profile Level 3 |
| Encoder          | MPEG-4 Simple Profile          |
|                  | H.263 Profile 0 Level 10-50    |
|                  | H.263 Profile 3 Level 10-50    |
|                  | H.264 Baseline Profile Level 3 |

***Decoding tools***

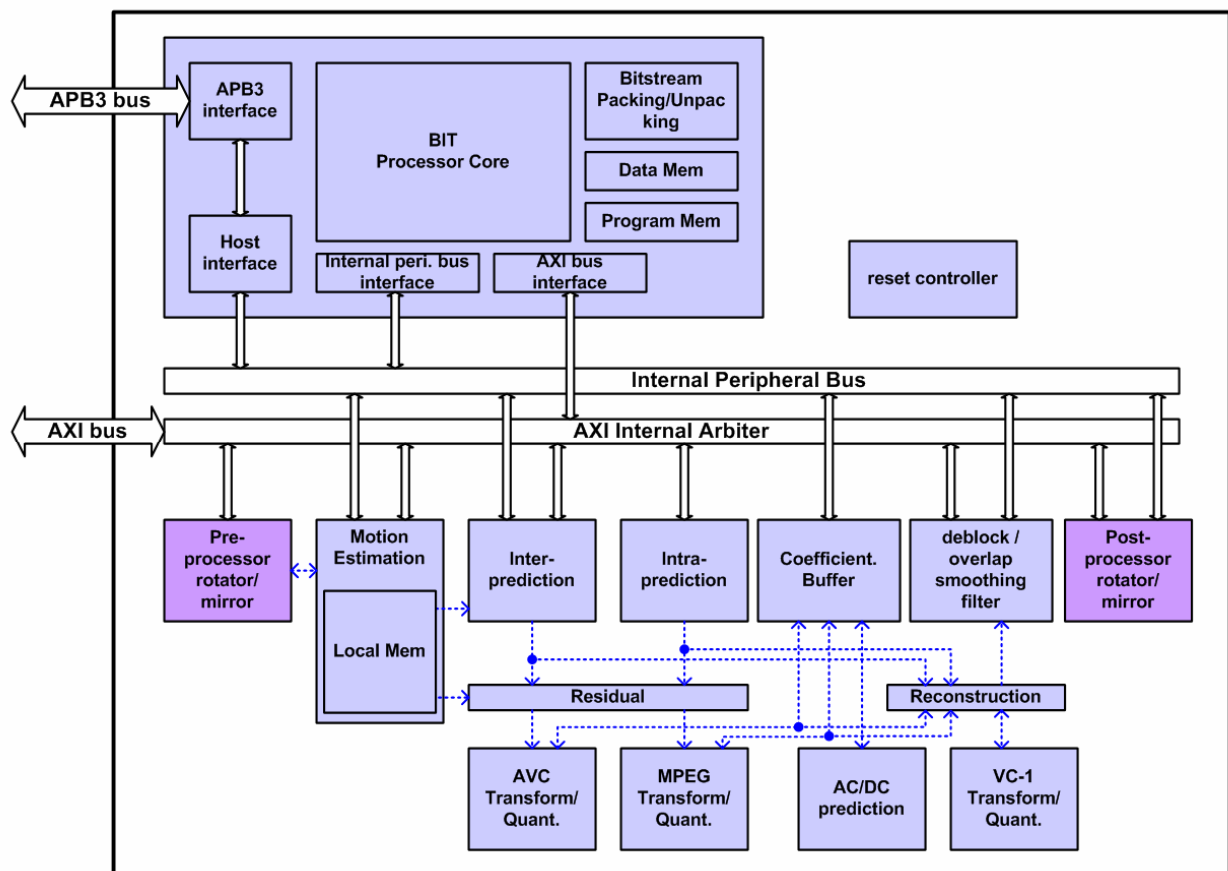
- MPEG-4 SP
  - 4MV with unrestricted motion vector
  - Error resilience tools: re-sync marker, data-partitioning with RVLC
- H.263 P3
  - P0 plus Annex I, J, K (RS=0 and ASO=0), and T supported
- H.264 BP
  - Variable block size supported (16x16, 16x8, 8x16, 8x8, 4x8, 4x4)
  -
- Minimum decoding size is 16 pixels in horizontal / 16-pixels in vertical

***Encoding tools***

- [+/-16,+/-16] 1/2 and 1/4-pel accuracy motion estimation
- 16x16, 16x8, 8x16, and 8x8 block sizes supported

- Block size is configurable
- Using only one reference frame for the motion estimation
- Unrestricted motion vector
- Prediction
  - MPEG-4 AC/DC prediction
  - H.264/AVC intra-prediction
  - H.263 Annex J, K (RS=0 and ASO=0), and T supported
- Error resilience tools
  - MPEG-4 re-sync. marker, data-partitioning with RVLC
- CIR (Cyclic Intra Refresh) / AIR (Adaptive Intra Refresh)
- Bit-rate control (CBR & VBR)
- Minimum encoding size is 32 pixels in horizontal / 16 pixels in vertical

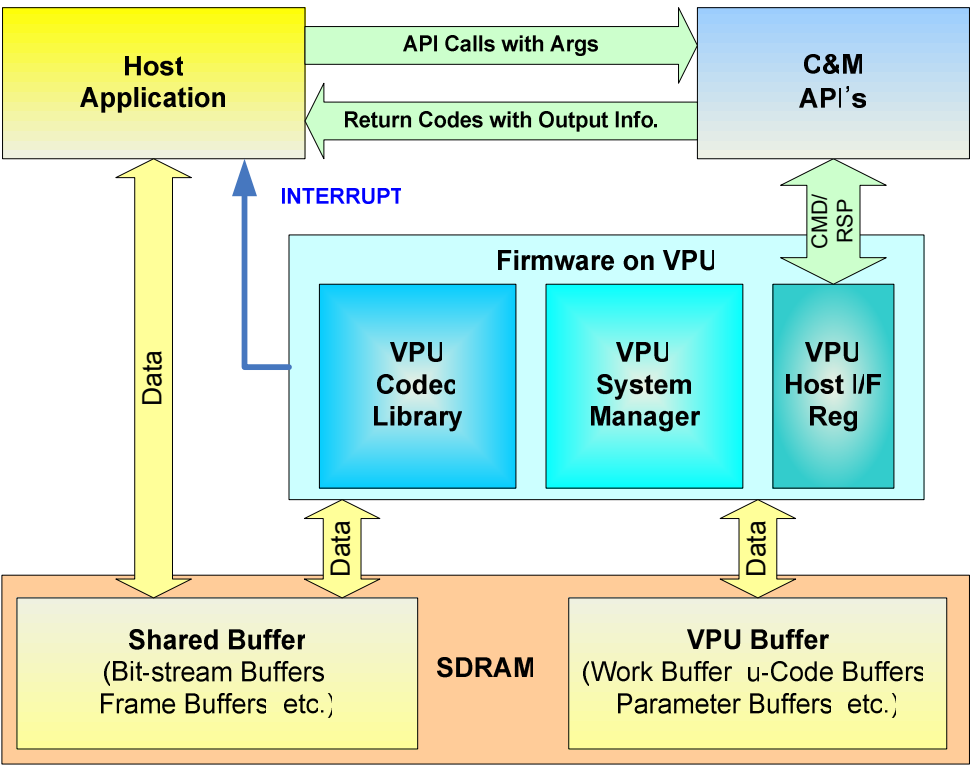
*General block diagram*



*BUS Architecture*

- CodaDx6 uses 2 external bus interfaces – APB, AXI and optional AHB to interface with user’s SoC, and internally the IPB (Internal Peripheral Bus).
- 32-bit APB interface (w/ PREADY)
    - interface between BPU and external host processor for configuring and reading status of video codec IP
  - 32-bit AXI interface
    - interface for reading data from and writing it to an external memory
  - Internal IPB bus
    - internal interface for communication between the BPU (or external host processor) and the video codec sub-modules

*Communication model*



All the bitstreams and picture data are directly accessed by host processor (e.g. ARM) and BPU(BIT Process Unit). The related

information for these data transfer will be exchanged through host interface by means of command and responses. For such operation, host interface of BPU reserves a set of registers accessible by host processor, some of which are used for exchanging actual command and responses. And others are used to provide to host processor such information as internal status of BPU.

A set of pre-defined commands and their corresponding responses are provided for use and reference. Customer-specific commands and responses could be easily added to a basic set by changing the definition of interface registers and a part of BIT processor code.

## Performance

All values are measured under **Typical case** using **PowerTheater** at RTL level.

### ■ Decoder performance (D1@30fps)

| <i>Video Standard</i> | <i>Max Bitrate<br/>(Mbps)</i> | <i>Core Clock<br/>(MHz)</i> | <i>Power Consumption<br/>(mW)</i> |
|-----------------------|-------------------------------|-----------------------------|-----------------------------------|
| MPEG-4 SP             | 4                             | 66                          | 84                                |
| H.263 P3              | 4                             | 66                          | 84                                |
| H.264 BP              | 3                             | 66                          | 95                                |

### ■ Encoder performance (D1@30fps)

| <i>Video Standard</i> | <i>Max Bitrate<br/>(Kbps)</i> | <i>Core Clock<br/>(MHz)</i> | <i>Power Consumption<br/>(mW)</i> |
|-----------------------|-------------------------------|-----------------------------|-----------------------------------|
| MPEG-4 SP             | 3                             | 70                          | 130                               |
| H.263 P3              | 3                             | 70                          | 130                               |

|          |   |    |     |
|----------|---|----|-----|
| H.264 BP | 4 | 70 | 130 |
|----------|---|----|-----|

- Performance information of CodaDx6 implemented with TSMC technology and FPGA are shown in the table

| <i>Technology</i> | <i>Gate count<br/>(Kgate)</i> | <i>Memory<br/>(Kbyte)</i> | <i>Area size<br/>(mm<sup>2</sup>)</i> |
|-------------------|-------------------------------|---------------------------|---------------------------------------|
| TSMC 0.13um       | 570                           | 34                        | 2.58x2.58                             |
| TSMC 90nm         | -                             | -                         | 1.78x1.78                             |
| Vertex LX200      |                               |                           |                                       |

*\* Area size is estimated including layout, routing, and clock tree.*

*\* Gate count does not include scan chain.*

### ***BUS Bandwidth requirement***

- Conditions:
  - D1@30fps with I-/P-/B-frames (**Typical case under lab condition**)
  - Read/Write latencies: 10/2
- Bandwidth required for decoder

| Standard  | Features     | Pure BandWidth |               | AXI          |               |
|-----------|--------------|----------------|---------------|--------------|---------------|
|           |              | read<br>MB/s   | write<br>MB/s | read<br>MB/s | write<br>MB/s |
| MPEG-4 SP | read / write | 27.01          | 19.78         | 35.76        | 20.70         |
|           | total        | 46.78          |               | 56.46        |               |
| H.264 BP  | read / write | 43.10          | 21.01         | 71.27        | 22.87         |
|           | total        | 64.12          |               | 94.14        |               |
| H.264 MP  | read / write | 69.13          | 22.87         | 113.67       | 25.03         |
|           | total        | 91.99          |               | 138.67       |               |

- Bandwidth required for encoder (w/o internal search RAM)

| Standard  | Features     | Pure BandWidth |               | AXI          |               |
|-----------|--------------|----------------|---------------|--------------|---------------|
|           |              | read<br>MB/s   | write<br>MB/s | read<br>MB/s | write<br>MB/s |
| MPEG-4 SP | read / write | 76.74          | 19.78         | 100.89       | 20.70         |
|           | total        | 96.52          |               | 121.59       |               |
| H.263 P3  | read / write | 76.74          | 19.78         | 102.44       | 21.01         |
|           | total        | 96.52          |               | 123.45       |               |
| H.264 BP  | read / write | 78.13          | 21.01         | 55.53        | 22.87         |
|           | total        | 99.14          |               | 78.39        |               |

- Bandwidth required for encoder (w/ internal search RAM: 27.3125KB)

| Standard  | Features     | Pure BandWidth |               | AXI          |               |
|-----------|--------------|----------------|---------------|--------------|---------------|
|           |              | read<br>MB/s   | write<br>MB/s | read<br>MB/s | write<br>MB/s |
| MPEG-4 SP | read / write | 37.66          | 19.78         | 49.50        | 20.70         |
|           | total        | 57.43          |               | 70.20        |               |
| H.263 P3  | read / write | 37.66          | 19.78         | 51.05        | 21.01         |
|           | total        | 57.43          |               | 72.06        |               |
| H.264 BP  | read / write | 39.05          | 21.01         | 55.53        | 22.87         |
|           | total        | 60.06          |               | 78.39        |               |

### ***Frame Buffer Requiriement***

| <i>Video Standard</i> | <i>Total frames</i> |
|-----------------------|---------------------|
| H.263 P3              | 6                   |
| MPEG-4 SP             | 6                   |
| H.264 BP              | 9                   |
| H.264 MP              | 9                   |



## Availability

**\* frames: reference + decoding + display + dbk + rotation**

**\* based on D1 resolution frame**

|                         | Status  |
|-------------------------|---------|
| FPGA proven             | Yes     |
| Test Chip Available     | Yes     |
| Licensed to Customers   | Yes     |
| Customer Silicon Proven | Yes     |
| In End Products         | Not yet |

## Key advantage features

### Ultra low-power design

It requires 15MHz core clock to decode H.264 CIF at 30fps.

### Multi-standard support

Any combination of four CIF decoders at 30fps is possible at 40MHz on FPGA.

IP in silicon is capable of decoding up to eight H.264 CIFs at 30fps or ten MPEG-4 CIFs at 30fps.

### Less than 1MIPs Host Processor required

CodaDx6 requires less than 1MIPs Host Processor, saving and reserving resources for other customers' purpose.

## Deliverables

Datasheet

Integration guide

Verification guide

RTL or Netlist w/ testbench

SW reference manual

Bit exact c-model



*Beyond the multimedia*

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